

Subject: All of Statistics, Random Variables

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1 Inequalities

Markov's Inequality and Chebyshev's Inequality

Theorem 1 (Markov's Inequality). 令 X 为随机变量, 假设 ① X 非负 ② $\mathbb{E}(X)$ 存在, 则

$$\mathbb{P}(X > t) \leq \frac{\mathbb{E}(X)}{t}, \quad \forall t > 0. \quad (1)$$

Proof. Due to $X > 0$, 因此

$$\begin{aligned} \mathbb{E}(X) &= \int_0^\infty xf(x)dx = \int_0^t xf(x)dx + \int_t^\infty xf(x)dx \\ &\geq \int_t^\infty xf(x)dx \geq \int_t^\infty tf(x)dx = t \mathbb{P}(X > t). \end{aligned} \quad (2)$$

□

Theorem 2 (Chebyshev's Inequality). 令 X 为随机变量, $\mu = \mathbb{E}(X)$, $\sigma^2 = \text{Var}(X)$, 则

$$P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}, \quad \forall t > 0 \quad \text{且} \quad P(|Z| \geq k) \leq \frac{1}{k^2}, \quad \forall k > 0, \quad (3)$$

其中 $Z = \frac{X - \mu}{\sigma}$.

Proof. 由于 $Z = \frac{X - \mu}{\sigma}$, 因此 $\mathbb{E}(Z) = 0$, $\text{Var}(Z) = 1$, $\mathbb{E}(Z^2) = \text{Var}(Z) + (\mathbb{E}(Z))^2 = 1$. 由 Markov's inequality 可得

$$P(|Z| \geq k) = P(Z^2 \geq k^2) \leq \frac{\mathbb{E}(Z^2)}{k^2} = \frac{1}{k^2} \Rightarrow P(|X - \mu| \geq t) = P(|Z| \geq \frac{t}{\sigma}) \leq \frac{\sigma^2}{t^2}. \quad (4)$$

□

Hoeffding's Inequality and Its Generalization

Theorem 3 (Hoeffding's Inequality). 设 $X_1, X_2, \dots, X_n$ 是 n 个独立的随机变量, 且 X_i 的取值范围为 $[a_i, b_i]$. 令 $S_n = \sum_{i=1}^n X_i$, 则对于任意 $t > 0$,

$$\begin{aligned} P(S_n - \mathbb{E}(S_n) \geq t) &\leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right), \\ P(|S_n - \mathbb{E}(S_n)| \geq t) &\leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right). \end{aligned} \quad (5)$$

Theorem 4 (Generalization of Hoeffding's Inequality). 设 $X_1, X_2, \dots, X_n$ 是 n 个独立的随机变量, $\mathbb{E}(X_i) = 0$ 且 X_i 的取值范围为 $[a_i, b_i]$. 令 $S_n = \sum_{i=1}^n X_i$, 则对于任意 $t, \epsilon > 0$,

$$P(S_n \geq \epsilon) \leq e^{-t\epsilon} \prod_{i=1}^n e^{t^2(b_i - a_i)^2/8}. \quad (6)$$

Theorem 5 (Generalization of Hoeffding's Inequality for Bernoulli). 令 $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, 则 $\forall \epsilon > 0$, 有

$$P(|\bar{X}_n - p| > \epsilon) \leq 2e^{-2n\epsilon^2}, \quad (7)$$

其中 $\bar{X}_n = \frac{\sum_{i=1}^n X_i}{n}$.

Proof. 可知 $\mathbb{E}(\bar{X}_n) = p$, 因此根据 Hoeffding's inequality Eq. (5) 有

$$P(|\bar{X}_n - p| > \epsilon) \leq 2e^{-\frac{2n^2\epsilon^2}{\sum_{i=1}^n(1-0)^2}} = 2e^{-2n\epsilon^2}. \quad (8)$$

□

Inequalities for Normal Distribution

Theorem 6 (Mill's Inequality). 令 $X \sim \mathcal{N}(0, \sigma^2)$, 则

$$P(|X| > t) \leq \sqrt{\frac{2}{\pi}} \frac{\sigma}{t} \exp\left(-\frac{t^2}{2\sigma^2}\right). \quad (9)$$

当 $\sigma^2 = 1$ $Z \sim \mathcal{N}(0, 1)$ 时满足

$$P(|Z| > t) \leq \sqrt{\frac{2}{\pi}} \frac{e^{-t^2/2}}{t}. \quad (10)$$

Inequalities For Expectations

Theorem 7 (Cauchy-Schwartz Inequality). 设 X, Y 为随机变量且方差有限, 即 $\text{Var}(X) < \infty, \text{Var}(Y) < \infty$, 那么

$$\mathbb{E}|XY| \leq \sqrt{\mathbb{E}(X^2)\mathbb{E}(Y^2)}. \quad (11)$$

Theorem 8 (Jensen's Inequality). 若函数 g 为凸函数 (*convex*), 则

$$\mathbb{E}g(X) \geq g(\mathbb{E}X). \quad (12)$$

函数 g 为凹函数 (*concave*), 则

$$\mathbb{E}g(X) \leq g(\mathbb{E}X). \quad (13)$$

A Proof

Proof for Hoeffding's Inequality

Lemma 1 (Hoeffding's Lemma). 设 X 是一个随机变量, 且 X 的取值范围为 $[a, b]$ (*almost surely*). 则对于任意 $\lambda > 0$, 有

$$\mathbb{E}(e^{\lambda X}) \leq \exp\left(\lambda \mathbb{E}(X) + \frac{\lambda^2(b-a)^2}{8}\right), \quad \text{or} \quad \mathbb{E}(e^{\lambda(X-\mathbb{E}(X))}) \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right). \quad (14)$$

Proof. 由于函数 $f(x) = e^{\lambda x}$ 是一个凸函数, 因此 X 的取值范围为 $[a, b]$ 可知, $\forall x \in [a, b]$, 有

$$e^{\lambda x} \leq \frac{b-x}{b-a} e^{\lambda a} + \frac{x-a}{b-a} e^{\lambda b}. \quad (15)$$

因此

$$\mathbb{E}(e^{\lambda X}) \leq \mathbb{E}\left(\frac{b-X}{b-a} e^{\lambda a} + \frac{X-a}{b-a} e^{\lambda b}\right) = \frac{b-\mathbb{E}(X)}{b-a} e^{\lambda a} + \frac{\mathbb{E}(X)-a}{b-a} e^{\lambda b}. \quad (16)$$

令 $p = \frac{\mathbb{E}(X)-a}{b-a}$, 显然 $p \in [0, 1]$, 此时有 $1-p = \frac{b-\mathbb{E}(X)}{b-a}$, $\mathbb{E}(X) = (1-p)a + pb$. 因此

$$\mathbb{E}(e^{\lambda X}) \leq (1-p)e^{\lambda a} + pe^{\lambda b} = e^{\lambda \mathbb{E}(X)} \left[(1-p)e^{\lambda(a-\mathbb{E}(X))} + pe^{\lambda(b-\mathbb{E}(X))} \right]. \quad (17)$$

由于 $\lambda(a-\mathbb{E}(X)) = \lambda(a-(1-p)a-pb) = -p\lambda(b-a)$, $\lambda(b-\mathbb{E}(X)) = \lambda(b-(1-p)a-pb) = (1-p)\lambda(b-a)$, 令 $u = \lambda(b-a)$ 得

$$(1-p)e^{\lambda a} + pe^{\lambda b} = e^{\lambda \mathbb{E}(X)} \left[(1-p)e^{-pu} + pe^{(1-p)u} \right]. \quad (18)$$

构造函数 $\phi(u) = \ln((1-p)e^{-pu} + pe^{(1-p)u})$, 可知

$$\begin{aligned} \phi(0) &= \ln((1-p)e^0 + pe^0) = \ln(1) = 0 \\ \phi'(u) &= \frac{-p(1-p)e^{-pu} + p(1-p)e^{(1-p)u}}{(1-p)e^{-pu} + pe^{(1-p)u}} \Rightarrow \phi'(0) = 0. \end{aligned} \quad (19)$$

令 $q(u) = \frac{pe^{(1-p)u}}{(1-p)e^{-pu} + pe^{(1-p)u}} \in [0, 1]$, 则

$$\phi'(u) = q(u) - p, \quad \phi''(u) = q'(u) = q(u)(1-q(u)) \leq \frac{1}{4}. \quad (20)$$

因此根据泰勒公式, $\exists \theta \in [0, u]$, 使得

$$\phi(u) = \phi(0) + \phi'(0)u + \frac{1}{2}\phi''(\theta)u^2 \leq \frac{u^2}{8}. \quad (21)$$

从而有

$$(1-p)e^{-pu} + pe^{(1-p)u} \leq \exp\left(\frac{u^2}{8}\right) \Rightarrow \mathbb{E}(e^{\lambda X}) \leq (1-p)e^{\lambda a} + pe^{\lambda b} \leq \exp\left(\lambda \mathbb{E}(X) + \frac{\lambda^2(b-a)^2}{8}\right). \quad (22)$$

当考虑 $\mathbb{E}(e^{\lambda(X-\mathbb{E}(X))})$ 时, 根据Eq. (17) 有

$$\mathbb{E}(e^{\lambda(X-\mathbb{E}(X))}) \leq (1-p)e^{\lambda a} + pe^{\lambda b} = \left[(1-p)e^{\lambda(a-\mathbb{E}(X))} + pe^{\lambda(b-\mathbb{E}(X))} \right] \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right). \quad (23)$$

□

下面证明 Theorem 3 Hoeffding's Inequality:

Proof. 根据 Markov's inequality 和 X_i 的独立性可知, $\forall \lambda, t > 0$:

$$P(S_n - \mathbb{E}(S_n) \geq t) = P(e^{\lambda(S_n - \mathbb{E}(S_n))} \geq e^{\lambda t}) \leq e^{-\lambda t} \mathbb{E}(e^{\lambda(S_n - \mathbb{E}(S_n))}) = e^{-\lambda t} \prod_{i=1}^n \mathbb{E}(e^{\lambda(X_i - \mathbb{E}(X_i))}). \quad (24)$$

由于 X_i 的取值范围为 $[a_i, b_i]$, 根据 Hoeffding's Lemma, 有 $\mathbb{E}(e^{\lambda(X_i - \mathbb{E}(X_i))}) \leq \exp\left(\frac{\lambda^2(b_i - a_i)^2}{8}\right)$, 从而有

$$P(S_n - \mathbb{E}(S_n) \geq t) \leq \exp\left(-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2\right). \quad (25)$$

根据二次函数的性质, $\forall t > 0$, 当 $\lambda = \frac{4t}{\sum_{i=1}^n (b_i - a_i)^2}$ 时, $-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2$ 取得最小值 $-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}$, 因此

$$P(S_n - \mathbb{E}(S_n) \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right). \quad (26)$$

同样地

$$P(S_n - \mathbb{E}(S_n) \leq -t) = P(e^{\lambda(\mathbb{E}(S_n) - S_n)} \geq e^{\lambda t}) \leq e^{-\lambda t} \mathbb{E}(e^{\lambda(\mathbb{E}(S_n) - S_n)}) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right). \quad (27)$$

综上所述, $\forall t > 0$, 有

$$P(|S_n - \mathbb{E}(S_n)| \geq t) = P(S_n - \mathbb{E}(S_n) \geq t) + P(S_n - \mathbb{E}(S_n) \leq -t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right). \quad (28)$$

□

下面证明 Theorem 4 Generalized Hoeffding's Inequality:

Proof. 根据函数 $f(x) = e^t x, t > 0$ 的单调性, 可知

$$P(S_n \geq \epsilon) = P(e^{tX_n} \geq e^{t\epsilon}). \quad (29)$$

根据 Markov's inequality 与 Hoeffding's lemma, 有

$$P(e^{tS_n} \geq e^{t\epsilon}) \leq e^{-t\epsilon} \mathbb{E}[e^{tS_n}] = e^{-t\epsilon} \prod_{i=1}^n \mathbb{E}[e^{tX_i}] \leq e^{-t\epsilon} \prod_{i=1}^n e^{t^2(b_i - a_i)^2/8}. \quad (30)$$

□

Proof of Mill's Inequality

Proof. 根据 Gaussian 的对称性, 有

$$P(|X| > t) = 2 \int_t^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} dx = \sqrt{\frac{2}{\pi}} \cdot \frac{1}{\sigma} \int_t^\infty e^{-\frac{x^2}{2\sigma^2}} dx < \sqrt{\frac{2}{\pi}} \cdot \frac{1}{t \cdot \sigma} \int_t^\infty x \cdot e^{-\frac{x^2}{2\sigma^2}} dx. \quad (31)$$

由于 $\left(-\sigma^2 e^{-\frac{x^2}{2\sigma^2}}\right)' = x e^{-\frac{x^2}{2\sigma^2}}$, 因此有

$$\sqrt{\frac{2}{\pi}} \cdot \frac{1}{t \cdot \sigma} \int_t^\infty x e^{-\frac{x^2}{2\sigma^2}} dx = \sqrt{\frac{2}{\pi}} \cdot \frac{1}{t \cdot \sigma} \left[-\sigma^2 e^{-\frac{x^2}{2\sigma^2}}\right]_t^\infty = \sqrt{\frac{2}{\pi}} \cdot \frac{1}{t \cdot \sigma} \left(0 - (-\sigma^2 e^{-\frac{t^2}{2\sigma^2}})\right) = \sqrt{\frac{2}{\pi}} \cdot \frac{\sigma}{t} e^{-\frac{t^2}{2\sigma^2}}. \quad (32)$$

因此有

$$P(|X| > t) \leq \sqrt{\frac{2}{\pi}} \frac{\sigma}{t} \exp\left(-\frac{t^2}{2\sigma^2}\right). \quad (33)$$

□

Proof of Cauchy-Schwartz Inequality

Proof. (Method I) ① 若 $\mathbb{E}(X^2) \cdot \mathbb{E}(Y^2) = 0$, 则显然 $\mathbb{E}(XY) = 0$, 故 $\mathbb{E}(|XY|) = 0$. ② 若 $\mathbb{E}(X^2) > 0$ 且 $\mathbb{E}(Y^2) > 0$, 考虑随机变量 $(|X| \cdot t + |Y|)^2$, 显然有

$$(|X| \cdot t + |Y|)^2 = |X|^2 t^2 + 2|XY|t + |Y|^2 = X^2 t^2 + 2|XY|t + Y^2 \geq 0. \quad (34)$$

因此有

$$\mathbb{E}[X^2 t^2 + 2|XY|t + Y^2] \geq \mathbb{E}[0] \Rightarrow \mathbb{E}(X^2) t^2 + 2\mathbb{E}|XY|t + \mathbb{E}(Y^2) \geq 0. \quad (35)$$

根据二次函数性质, 由于 $\mathbb{E}(X^2) > 0$, 因此

$$(2\mathbb{E}|XY|)^2 - 4 \cdot \mathbb{E}(X^2) \cdot \mathbb{E}(Y^2) \leq 0 \Rightarrow \mathbb{E}|XY| \leq \sqrt{\mathbb{E}(X^2)\mathbb{E}(Y^2)}. \quad (36)$$

(Method II) 构造新随机变量 $U = \frac{|X|}{\sqrt{\mathbb{E}(X^2)}}$, $V = \frac{|Y|}{\sqrt{\mathbb{E}(Y^2)}}$, 此时显然有 $\mathbb{E}(U^2) = 1, \mathbb{E}(V^2) = 1$, 根据不等式 $UV \leq \frac{U^2 + V^2}{2}$ 有:

$$\mathbb{E}(UV) \leq \frac{\mathbb{E}(U^2) + \mathbb{E}(V^2)}{2} = \frac{1 + 1}{2} = 1 \Rightarrow \mathbb{E}|XY| \leq \sqrt{\mathbb{E}(X^2)\mathbb{E}(Y^2)}. \quad (37)$$

□

Proof of Jensen's Inequality

Proof. 由于 $g(x)$ 为凸函数, 根据凸函数的性质, 在点 $(\mu, g(\mu))$ 处, 必然存在一条支撑线, 设其斜率为 L , 则这条直线的方程可以表示为:

$$y = g(\mu) + L(x - \mu). \quad (38)$$

根据支撑线的定义, 函数图像上的任何点都位于这条直线的上方或其上:

$$g(x) \geq g(\mu) + L(x - \mu), \quad \forall x \Rightarrow g(X) \geq g(\mu) + L(X - \mu). \quad (39)$$

根据期望的单调性和线性性质, 有

$$\mathbb{E}[g(X)] \geq \mathbb{E}[g(\mu) + L(X - \mu)] = \mathbb{E}[g(\mu)] + L \cdot \mathbb{E}[X - \mu] = g(\mathbb{E}X). \quad (40)$$

当 $g(x)$ 为凹函数时, 考虑 $h(x) = -g(x)$ 为凸函数即可. $\square$

Proposition 1. 若 X 为只取有限值 $x_1, x_2, \dots, x_n$ 的离散随机变量, 对应的概率为 $p_1, \dots, p_n$ (满足 $p_i \geq 0$ 且 $\sum_{i=1}^n p_i = 1$), g 为凸函数(满足 $g(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda g(x_1) + (1 - \lambda)g(x_2)$), 则:

$$g\left(\sum_{i=1}^n p_i x_i\right) \leq \sum_{i=1}^n p_i g(x_i). \quad (41)$$

Proof. 使用数学归纳法. ① 当 $n = 1$ 时: $p_1 = 1$, $g(x_1) \leq g(x_1)$ 显然成立. ② 当 $n = 2$ 时: 令 $\lambda = p_1$, $1 - \lambda = p_2$, 根据凸函数的定义, 自然成立. ③ 假设当 $n = k$ 时命题成立, 即

$$g\left(\sum_{i=1}^k p_i x_i\right) \leq \sum_{i=1}^k p_i g(x_i), \quad \sum_{i=1}^k p_i = 1. \quad (42)$$

当 $n = k + 1$ 时, 有

$$\sum_{i=1}^{k+1} p_i x_i = (1 - p_{k+1}) \left(\sum_{i=1}^k \frac{p_i}{1 - p_{k+1}} x_i \right) + p_{k+1} x_{k+1}. \quad (43)$$

由于 $(1 - p_{k+1}) + p_{k+1} = 1$, 根据 $n = 2$ 时结论有

$$g\left(\sum_{i=1}^{k+1} p_i x_i\right) \leq (1 - p_{k+1}) \cdot g\left(\sum_{i=1}^k \frac{p_i}{1 - p_{k+1}} x_i\right) + p_{k+1} g(x_{k+1}). \quad (44)$$

由于 $\frac{p_i}{1 - p_{k+1}} = 1$, 根据 $n = k$ 时结论有

$$g\left(\sum_{i=1}^k \frac{p_i}{1 - p_{k+1}} x_i\right) \leq \sum_{i=1}^k \frac{p_i}{1 - p_{k+1}} g(x_i). \quad (45)$$

从而有

$$g\left(\sum_{i=1}^{k+1} p_i x_i\right) \leq (1 - p_{k+1}) \cdot \left[\sum_{i=1}^k \frac{p_i}{1 - p_{k+1}} g(x_i) \right] + p_{k+1} g(x_{k+1}) = \sum_{i=1}^{k+1} p_i g(x_i). \quad (46)$$

根据归纳, 可得

$$g\left(\sum_{i=1}^n p_i x_i\right) \leq \sum_{i=1}^n p_i g(x_i), \quad \sum_{i=1}^n p_i = 1. \quad (47)$$

$\square$

Proposition 2. 设 X 为一连续型随机变量, g 为凸函数, 则

$$g(\mathbb{E}X) \leq \mathbb{E}g(X). \quad (48)$$

Proof. 对于随机变量 X ，可以构造一个简单随机变量序列 $\{X_m\}_{m=1}^{\infty}$ ，其中每一 X_m 都为只取有限个值的离散变量，使得：

$$\lim_{m \rightarrow \infty} X_m = X. \quad (49)$$

根据勒贝格控制收敛定理，有

$$\lim_{m \rightarrow \infty} \mathbb{E}(X_m) = \mathbb{E}(X). \quad (50)$$

由于每个 X_m 都是只取有限个值的离散变量，根据Proposition 1，有

$$g(\mathbb{E}(X_m)) \leq \mathbb{E}[g(X_m)], \quad \forall m. \quad (51)$$

对上式两边同时取 $m \rightarrow \infty$ 的极限，有

$$\begin{aligned} (LHS) \quad \lim_{m \rightarrow \infty} g(\mathbb{E}(X_m)) &= g\left(\lim_{m \rightarrow \infty} \mathbb{E}(X_m)\right) = g(\mathbb{E}(X)), \\ (RHS) \quad \lim_{m \rightarrow \infty} \mathbb{E}[g(X_m)] &= \mathbb{E}\left[\lim_{m \rightarrow \infty} g(X_m)\right] = \mathbb{E}[g(X)]. \end{aligned} \quad (52)$$

从而得到 $g(\mathbb{E}X) \leq \mathbb{E}g(X)$. □